Customer Churn and Retention Analysis for a Subscription-Based Business

Introduction:

Customer churn refers to the percentage of customers who stop using a service over a given period. In subscription-based businesses, high churn rates directly affect revenue and long-term growth.

This project aims to analyze customer data to identify churn patterns, understand the factors influencing customer retention, and evaluate customer lifetime trends. The insights obtained from this analysis will help businesses make data-driven decisions to reduce churn and improve customer retention.

Dataset Description:

The dataset used in this analysis is the Telco Customer Churn dataset, which contains information about customers and their subscription behavior.

Each row represents a customer, and each column describes specific attributes such as:

- Demographic information (gender, senior citizen status, dependents)
- Account details (tenure, contract type, payment method)
- Services subscribed (internet service, online security, tech support, etc.)
- Financial data (monthly charges, total charges)
- Churn status (whether the customer left or stayed)

The dataset enables analysis of customer behavior and identification of factors contributing to churn.

Data Cleaning and Preparation:

Before analysis, the dataset was cleaned and prepared to ensure accuracy and consistency.

The following steps were performed:

- The TotalCharges column was converted to a numeric data type and missing values were handled.
- A new column called Churn_Flag was created to convert churn values into numeric form (1 = Yes, 0 = No).
- A Customer Lifetime Value (CLV) column was created by multiplying monthly charges by tenure.
- A Tenure_Group column was created to categorize customers into groups based on how long they have been subscribed (0–12, 13–24, 25–48, 49+ months).

These transformations enabled effective analysis and visualization in Power BI.

Churn Analysis:

The analysis revealed several key patterns in customer churn.

Customers on month-to-month contracts have the highest churn rate compared to those on one-year and two-year contracts. This indicates that customers without long-term commitments are more likely to leave.

Customers within the first 0–12 months show the highest churn rate, suggesting that early customer experience plays a critical role in retention.

Additionally, customers using electronic check payment methods exhibit higher churn rates compared to other payment methods, indicating possible inconvenience or dissatisfaction with this payment option.

Retention Drivers:

The analysis of service-related variables revealed important retention drivers.

Customers without technical support services show significantly higher churn rates compared to those who have access to support. This suggests that technical assistance improves customer satisfaction and retention.

Similarly, customers without online security and backup services tend to churn more, indicating that value-added services play a role in keeping customers engaged.

Overall, customers who subscribe to additional services are more likely to remain with the company.

Customer Lifetime Trends:

Customer lifetime analysis shows that customers with longer tenure contribute more value to the business.

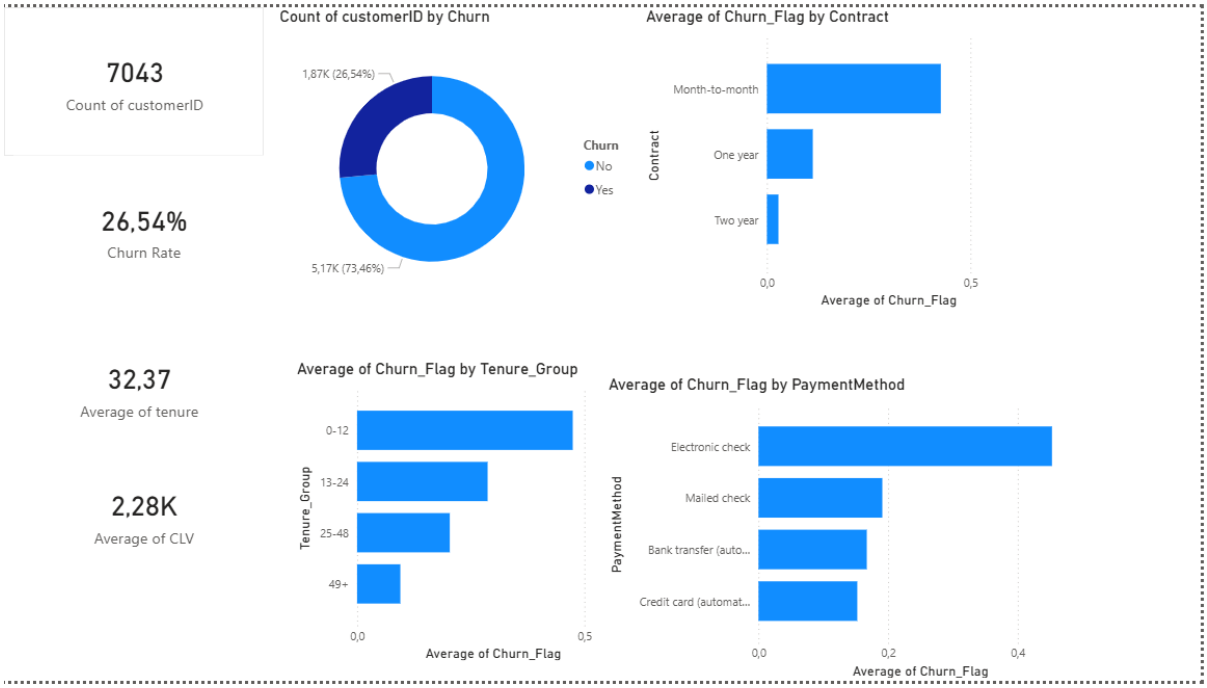
Customers who remain subscribed for longer periods generate higher Customer Lifetime Value (CLV), while those who churn early contribute significantly less.

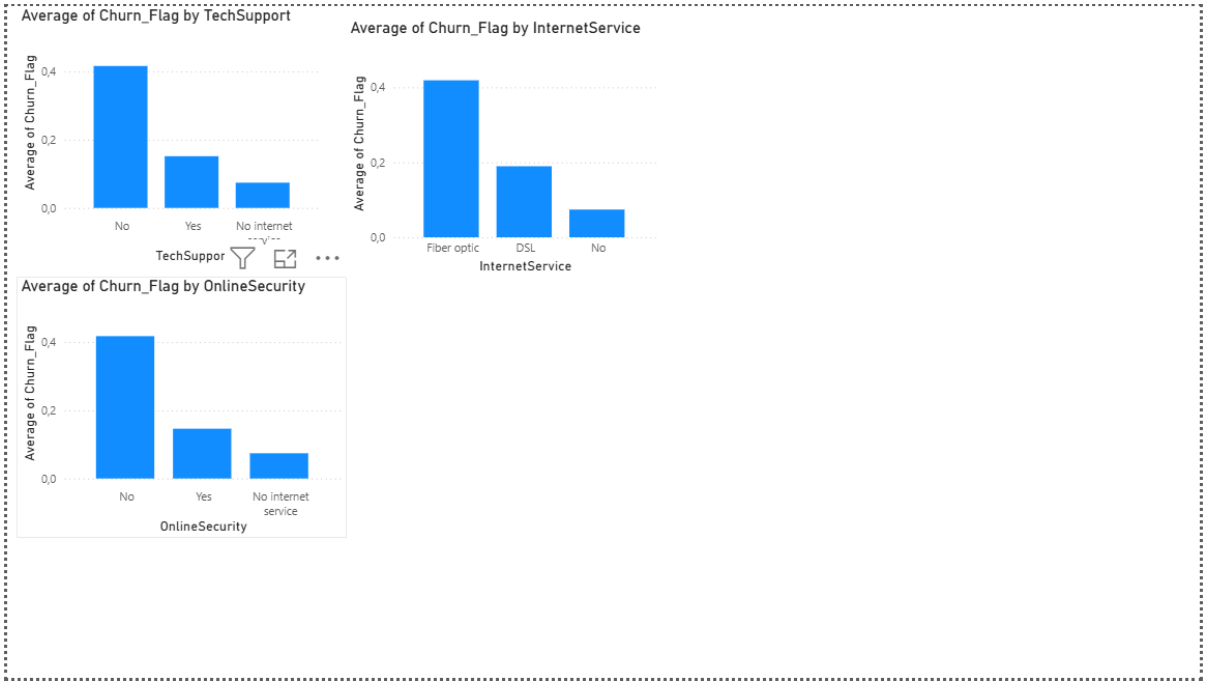
This highlights the importance of retaining customers beyond the initial stages of their subscription.

Based on the analysis, the following recommendations are proposed:

- Improve customer onboarding processes to reduce churn in the first 12 months.
- Encourage customers to switch to long-term contracts through incentives such as discounts.
- Promote value-added services such as technical support, online security, and backup.
- Improve payment experience and encourage more convenient payment methods.
- Monitor high-risk customers and implement targeted retention strategies.

These actions can help reduce churn and improve overall customer retention.

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TechSuppor
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Average of Churn_Flag
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0,2
0,0
Fiber optic
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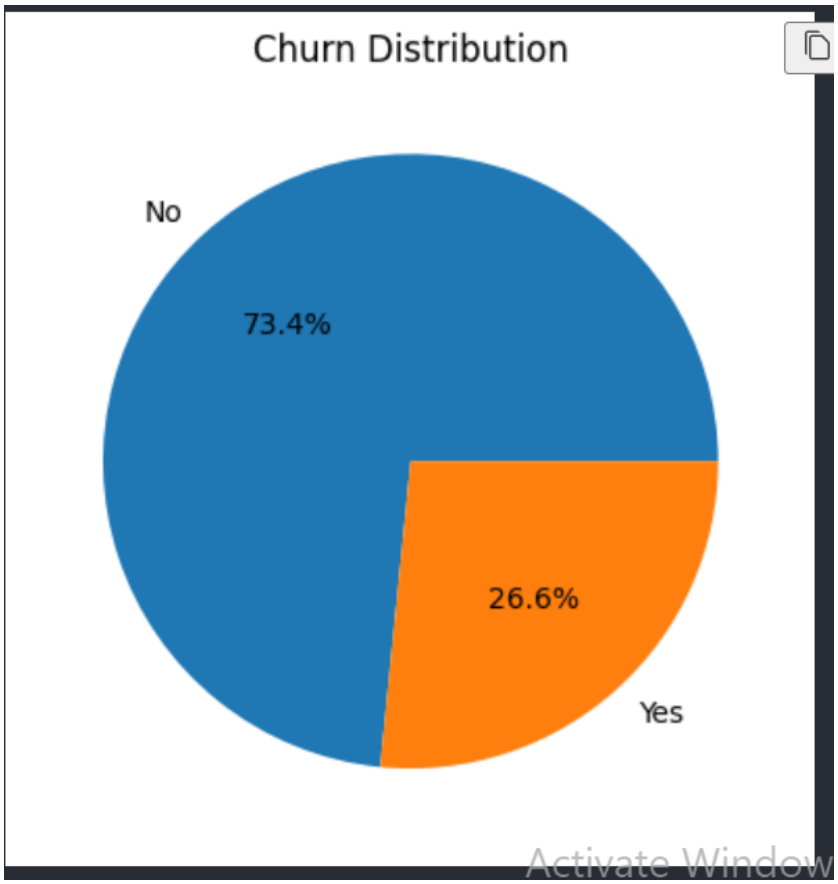
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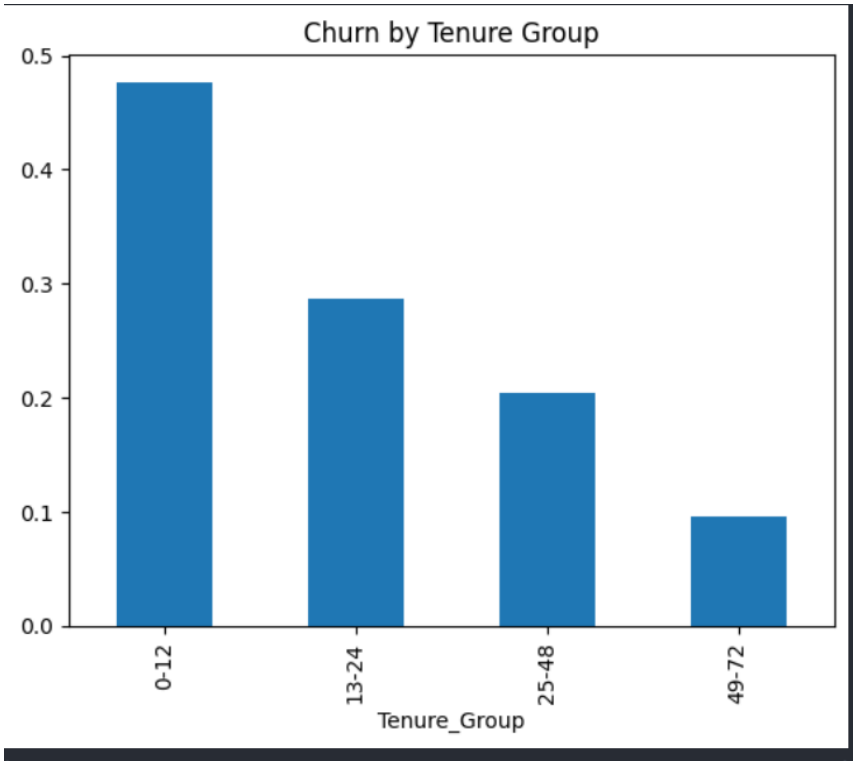
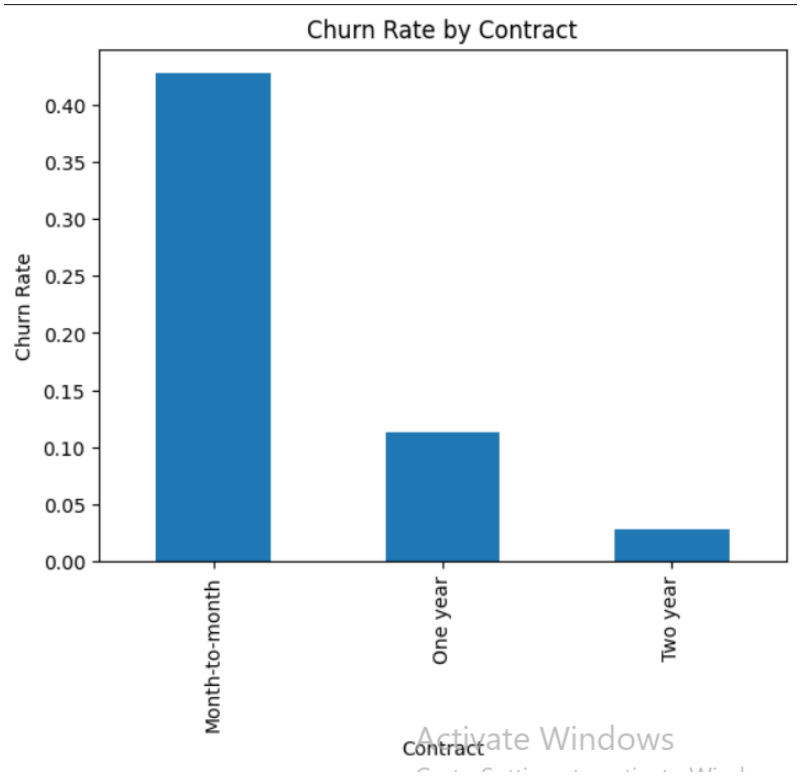
Activate Window





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